

Abdul-Kazeem Shamba

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🤖 Robotic Engineer
🧠 Artificial Intelligence Researcher
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Doctoral Researcher in Computer Science at NTNU with expertise in representation learning, computer vision, and multimodal sensor data for robotics applications. Holds four advanced degrees across AI, robotics, and engineering, including an MSc in Marine Intelligent Robotics and ECE from Carnegie Mellon. Experienced in developing full training pipelines, deploying real-world robotics and vision systems, and publishing in top AI conferences. Strong background in self-supervised and contrastive learning, satellite data visualization, and autonomous systems, with hands-on skills in Python, PyTorch, ROS, C++, Docker, and embedded control.

EDUCATION

Doctor of Philosophy (PhD) in Computer Science , <i>Norwegian University of Science & Tech., Norway</i>	2023 — Present
Master of Science in Marine Intelligent Robotics , <i>Norwegian University of Science & Tech., Norway</i>	2022 — 2023
Master of Science in Marine and Maritime Intelligent Robotics , <i>Univeristy of Toulon, France</i>	2021 — 2022
Master of Science in Electrical and Computer Engineering , <i>Carnegie Mellon University, USA</i>	2020 — 2021
Bachelor of Science in Electronic and Electrical Engineering , <i>Obafemi Awolowo University, Nigeria</i>	2013 — 2018

- **PhD's Tentative Thesis:** "Representation Learning for Predictive Analytics and Reasoning in Multimodal Sensor Data"
- **Master's Thesis:** "Generating Realistic Underwater Renderings using Generative Adversarial Networks" - Grade A
- **Bachelor's Thesis:** "Design and Development of an Arm Prosthesis using Electromyogram Control" - Grade A
- **Standardize Test Scores:** GRE 317 (Q162 V155 AWA4) & TOEFL 101 (R27 L27 S22 W25)

Top Courses: Intelligent Agents, Probabilistic Artificial Intelligence, Visual Computing (3D Graphics Rendering), Autonomous Marine Systems, Introduction to Visual SLAM, Artificial Intelligence, Advanced Control of Auto. System, Data Inference and Applied ML, ML for Signal Proc., Applied Computer Vision, Natural Language Proc., Introduction to Deep Learning, Intro. to Cognitive Robotics, Big Data Science, Microelectronic Devices and Circuits, Digital Circuit Analysis and Design, Nanotechnology and Nanoelectronics

PROJECT EXPERIENCE

- Designed an arm prosthesis using an Electromyogram signal and executed with python programming language and C on the Arduino interface to minimize the constraints embedded in traditional static prosthesis by over 50%
- Developed a speech-to-text model with an attention mechanism from scratch for video captioning using CNNs and bi-directional LSTMs and achieved a Levenshtein distance of 17 to qualify for an A grade and top 11% in Kaggle

TECHNICAL EXPERIENCE

DOCTORAL RESEARCHER / CONTRASTIVE REPRESENTATION LEARNING <i>Norwegian University of Science and Technology</i>	Sep 2023 — Present <i>Trondheim, Norway</i>
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- Published multiple works on contrastive representation learning in top AI conference with less than 25% acceptance rate
- Contributed a total of 782 hours as a teaching assistant and oral exam sensor across five courses in computer science
- Participated in talks on meta-learning and contrastive representation learning in AI conferences and workshops in Europe

COMPUTER VISION INTERN / SELF-SUPERVISED LEARNING RESEARCH <i>Alteia</i>	Jun 2022 — Aug 2023 <i>Toulouse, France</i>
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- Developed and coded the entire training and evaluation pipeline on 3 different pretext tasks from scratch
- Improved the overall accuracy of a vanilla supervised model by 6.5% after pretraining via self-supervised learning
- Visualized the model's results and evaluation metrics on the WandB platform by refactoring the entire code

GRADUATE TEACHING ASSISTANT / ADVANCED COMPUTER VISION <i>Carnegie Mellon University</i>	Aug 2021 — Dec 2021 <i>Kigali, Rwanda</i>
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- Reduced assignment grading time by 40% through writing scripts and python codes to grade individual submissions
- Facilitated weekly tutorials for 15 graduate students on homework and difficult concepts covered in the classroom
- Graded 50% of the assignments and computed the class distributions for all 15 students in the fall 2021 semester

SOFTWARE DEVELOPER / SATELLITE DATA VISUALIZATION WEB-DASHBOARD <i>Technoserve</i>	May 2021 — Nov 2021 <i>Cotonou, Benin</i>
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- Developed and deployed a web dashboard for satellite data visualization of cashew farms using Python
- Integrated ground data statistics of 1900 plantations in Benin republic with the shapefiles on the web dashboard

VOLUNTEER EXPERIENCE

STUDENT AMBASSADOR / FACULTY OF ENGINEERING

Norwegian University of Science and Technology

Sep 2022 — Present

Trondheim, Norway

- Create stories for IV's study program pages (Gobi stories) and social media
- Present the study at school visits and campuses for both physical and digital events

DIRECTOR OF IT COMMUNICATIONS / INT'L STUDENT UNION TRONDHEIM

Norwegian University of Science and Technology

Sep 2022 — Present

Trondheim, Norway

- Organized quizzes and games events for over 200 international students in Trondheim during the 2022/2023 session
- Presented the organization's objectives and activities to about 500 new international students during the NTNU spring 2023 orientation

ADVOCATE / GLOBAL SCHOOLS PROGRAM

Non-Profit Organization

Aug 2019 — Feb 2020

Niger, Nigeria

- Created awareness about the SDGs and helped schools and educators become engaged with it
- Provided oral presentations about the relevance of the SDGs to school authorities, educators, and students

TEACHER / DEPARTMENT OF ENGLISH

Government Technical College

Mar 2019 — Feb 2020

New Bussa, Nigeria

- Taught 198 Technical College students English Language in Niger State for a minimum of 3 hours per week
- Marked and supervised assignments, tests, and exams alongside organizing a weekly career fair and counseling talk

WARD ACCOUNTABILITY OFFICER / MALALE WARD

Borgu Primary Health Care Center,

Apr 2019 — Jun 2019

Kainji, Nigeria

- Ensured the Oral Poliomyelitis Vaccine (OPV) carriers were labeled with MOPV (Monovalent Oral Poliovirus Vaccine)
- Attended daily review meetings with the primary health care director, WHO representatives and ward focal persons

TECHNICAL SKILLS

Tools and Languages	Python, Git, $\LaTeX$, C++, Java, OpenGL, Heroku, JavaScript, Flask, Django, Linux, Docker, MySQL, GLSL, Rust, CuDA, ROS
Data Visualization	Jupyter, Numpy, Matplotlib, Pandas, OpenCV, Blender, Google Earth, Folium, Scikit-learn
Machine Learning	Gaussian Mixture Models, Hidden Markov Models, K-Means, Random Forest, PCA, NLTK
Deep Learning	Multilayer Perceptron, Convolution Neural Network, Recurrent & GAN Network, Pytorch

RESEARCH WORK

1. Shamba, A. "Self-Supervised Learning: In-depth Performance Analysis of Different Pretext Tasks on Established Benchmarks", Alteia Summer Internship, Toulouse, France, 2022
2. Shamba, A. "Fish Behavior Analysis in Video Sequence using Deep Bi-Directional LSTM with CNN Features", Poster Presentation, Toulon, France, MIR 2022 Symposium
3. Shamba, A. "Satellite Mapping and Visualization of Cashew Plantations in Nigeria", Nile University of Nigeria Master's Thesis, Abuja, Nigeria, 2022
4. Shamba, A., Akanji, D. "Anomaly Detection in Video Surveillance using Neural Networks and Naive Bayes Classifier", Deep Learning Course, University of Toulon, France, 2021
5. Shamba, A., Tinuoye, J. "Design and Development of 3D Arm Prosthesis using EMG Control", Obafemi Awolowo University Bachelor's Thesis, Ile-Ife, Nigeria, 2018

AWARDS AND ACHIEVEMENT

Awarded Best Poster at NorwAI Innovate Conference, Trondheim Norway	Sept. 2024
Won the Erasmus Mundus Joint Masters Scholarship, France and Norway	June 2021
Best Lagos state candidate in the National Information Technology Development scholarship, Nigeria	Oct. 2020
Awarded the Carnegie Institute of Technology Dean's Fellowship, Pittsburgh, USA	June 2020
Secured the MTN telecommunication undergraduate foundation scholarship, Nigeria	March 2014
Graduated Overall best at Yaro College Secondary School, Nigeria	Sept. 2010

LANGUAGES

English (Native proficiency), **Hausa** (Limited working proficiency), **French**, **Yoruba** and **Arabic** (Elementary proficiency)